QUESTION :
Design a “farm animals” java application with the details of animals like cow, pig, horse. Consider the following details like where they stay, what they eat, the sound they make by using classes and objects.

EXECUTION STEPS :

- **Open IntelliJ IDEA** and create a new Java project with the required JDK.
- **Create a Java class** named `FarmAnimals` and write the program using classes and objects for **Cow, Pig, and Horse**.
- **Compile the program** and check that there are no syntax errors.
- **Run the program** using the green **Run** button in IntelliJ IDEA.
- **View the output** in the IntelliJ console. The program displays where each animal stays, what it eats, and the sound it makes.

ALGORITHM :

- **Start** the program.
- Create a class named `Animal` with the attributes:
  - `name`
  - `stay`
  - `food`
  - `sound`
- Create a **constructor** in the `Animal` class to initialize the details of each animal.
- Create a `display()` method to display the animal's name, where it stays, what it eats, and the sound it makes.
- In the `main()` method, create objects for:
  - **Cow**
  - **Pig**
  - **Horse**
- Initialize the details of each animal using the constructor.
- Call the `display()` method for each object.
- Display the details of the cow, pig, and horse.
- **Stop** the program.

UML DIAGRAM :

```
┌──────────────────────────┐
│        Animal            │
├──────────────────────────┤
│ - name : String          │
│ - stay : String          │
```

```
| - food : String              |
| - sound : String             |
├──────────────────────────────┤
| + Animal(String,String,      |
|   String,String)             |
| + display() : void           |
└──────────────────────────────┘

      ▲
      |
Objects created from
   Animal class
  /    |    \
 /     |     \
▼      ▼      ▼

┌──────┐ ┌──────┐ ┌────────┐
| cow  | | pig  | | horse  |
└──────┘ └──────┘ └────────┘
```

CODE :

```
import java.util.Scanner;
class Animal{
	void stay() {
		System.out.println("animals do live outside");
	}
	void food() {
		System.out.println("animmals eat food");
	}
	void sound() {
		System.out.println("animals do sounds");
	}
}
class Cow extends Animal{
	void stay(){System.out.println("cows live in the farm");
	}
	void food() {
		System.out.println("cows eat grass");
	}
	void sound() {
		System.out.println("cows do a sounds like Mooo...");
	}
}
class Pig extends Animal{
	void stay(){System.out.println("pigs live in the bushes");
	}
	void food() {
		System.out.println("pigs eat  fruits,vegitables");
	}
```

```
        void sound() {
                System.out.println("pigs do a sounds like oink...");
        }
}
class Horse extends Animal{
        void stay(){System.out.println("horse live in the horsefarm");
        }
        void food() {
                System.out.println("horse eat grass,hay");
        }
        void sound() {
                System.out.println("horse do a sounds called whinny...");
        }
}
public class AnimalFarm {
public static void main(String[] args) {
        Cow c=new Cow();
        Pig p=new Pig();
        Horse h=new Horse();
        System.out.println("the cow,pig,horse details are shown below");
        c.stay();
        c.food();
        c.sound();
        p.stay();
        p.food();
        p.sound();
        h.stay();
        h.food();
        h.sound();

}
}
```

OUTPUT :

```
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=51338" -Dfile.encoding=UTF-8
the cow,pig,horse details are shown below
cows live in the farm
cows eat grass
cows do a sounds like Mooo...
pigs live in the bushes
pigs eat  fruits,vegitables
pigs do a sounds like oink...
horse live in the horsefarm
horse eat grass,hay
horse do a sounds called whinny...

Process finished with exit code 0
```